

Joshua Dinn

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Education

Monash University, Melbourne, Australia

Mar 2023 – Oct 2027

Bachelor of Engineering (Honours, Software Engineering) and Bachelor of Commerce (Econometrics)

WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

Exchange: University of Copenhagen (Sem 1 2025), Econometrics and Databases & Information Systems

Experience

Aumont, Founding Software Engineer

Jul 2026 – Present

- Help small and mid-sized businesses automate manual processes, cut operating costs and modernise internal software, using agentic AI tooling to ship tools in days rather than weeks.
- Design SDLC and release processes for clients deploying straight to production: a staging environment with masked production data, deliberate database migrations in place of ad-hoc schema changes on boot, enforced merge and release gates, and monitoring across scheduled jobs and third-party integrations.
- Run security reviews that close live vulnerabilities and rebuild login and access controls, and automate manual data ingestion (supplier feeds, stock updates) end to end.

Collingwood Football Club, Data, Analytics and Technology Intern

Mar 2026 – Jun 2026

- Built a membership churn prediction model in XGBoost, engineering feature pipelines from historical transaction and engagement data to flag at-risk members for the commercial team.
- Built a SHAP dashboard of churn drivers, giving stakeholders a self-serve view of retention risk.
- Designed a repository synchronisation workflow between the club's web and mobile codebases, removing a manual step from every deployment.

Monash DeepNeuron, Optimised Computing Member

Sep 2025 – Present

- HarvestAI: building the perception module for an autonomous strawberry-picking arm, training an object-detection model on HPC clusters to find strawberries in a live camera feed and classify ripeness.
- Built Omni AR, a smart-glasses assistant with photographic memory: Gemini 2.5 Flash for vision and embeddings, a local vector database for semantic recall, and Whisper and TTS streamed over SSE for low latency.
- Teaching Assistant for an 8-week C++ and accelerated computing workshop on GPU programming, CUDA and HPC, run with NVIDIA, Pawsey Supercomputing Centre and NCI. NVIDIA CUDA certified.

Monash eSolutions, AI Fitness Trainer

Jul 2026 – Present

- Run in-person AI zones for students and staff, handling everything from first-time questions to technical queries on Monash's enterprise AI tools. Delivered two university webinars on AI, 200+ attendees each.

Projects

JackTen: No-Limit Hold'em Poker Bot | Python, Bayesian inference, LSTM

In progress

- Building a strategy engine from the math up: hand evaluation and equity, then the EV, pot odds and minimum-defence-frequency formulas behind every decision. Currently through the equity and EV layers.
- Next: Bayesian range updates from opponent stats (VPIP, PFR, 3-bet%), fed by LSTM sequence models trained on hand histories to predict opponent tendencies.

StdOut: Real-Time AI Technical Interview Simulator | React, Node.js, MongoDB, WebSockets

Feb 2026

- Built a real-time interview platform with a 3-person hackathon team, deployed on Vercel, Render and Atlas.
- Designed the architecture to synchronise candidate and interviewer state over WebSockets, keeping transcription, AI responses and code consistent, with streaming handled asynchronously so it never blocked the session.
- Built a Python execution terminal that tracks code diffs as events, enabling full replay of a candidate's session.

Maze Solver & Algorithm Visualiser | Python, pygame, Hypothesis, pytest

Oct 2025

- Built a pathfinding visualiser with play, watch and race modes across 7 algorithms on procedurally generated mazes.
- Tested it with property-based tests in Hypothesis and visual snapshot tests, and benchmarked 125,000+ simulated runs. Major project for FIT2107 (100%).

Algorithmic Trading System for S&P 500 ETF | Python, HMM, AWS Lambda, Alpaca API

May 2025

- Built a regime-switching HMM strategy with RSI, ATR and volatility-based sizing, paper trading on AWS Lambda.

Skills

Languages: Python, C++, TypeScript, JavaScript, Java, SQL, MATLAB, R

Frameworks & Tools: React, Node.js, Flask, MongoDB, AWS, Linux, Git, pandas, NumPy

AI & ML: PyTorch, XGBoost, scikit-learn, SHAP, LLM APIs, RAG, vector databases, CUDA